

Chapter 7

Emerging Trends

Overview

7.1 Sequence to Sequence Models

7.2 Federated Learning

7.3 Edge AI

7.4 Ethics and AI: Responsible AI

7.1 Sequence to Sequence Models

- If we want to build Google Translate which converts sentences in one language to another language such as Input: "I love Nepal." & Desired output: "म नेपाललाई माया गर्छु।"
- Here the length of inputs is 3 words while output length is 4 words.
- Traditional neural networks cannot naturally handle variable-length input and output. We need a new architecture. This leads to the Sequence-to-Sequence Model.
- A sequence is simply an ordered list of elements.
- Examples: Sentence: "I love AI"
- Sequence is considered as Word1 → Word2 → Word3.
- A Sequence-to-Sequence model converts one sequence into another sequence.

7.1 Sequence to Sequence Models

- Sequence-to-Sequence (Seq2Seq) models are neural networks designed to transform one sequence into another, even when the input and output lengths differ and are built using encoder-decoder architecture.
- It processes an input sequence and generates a corresponding output sequence.
- Handles variable-length input and output sequences
- It is used in NLP, machine translation, speech recognition and time-series prediction.

7.1 Sequence to Sequence Models

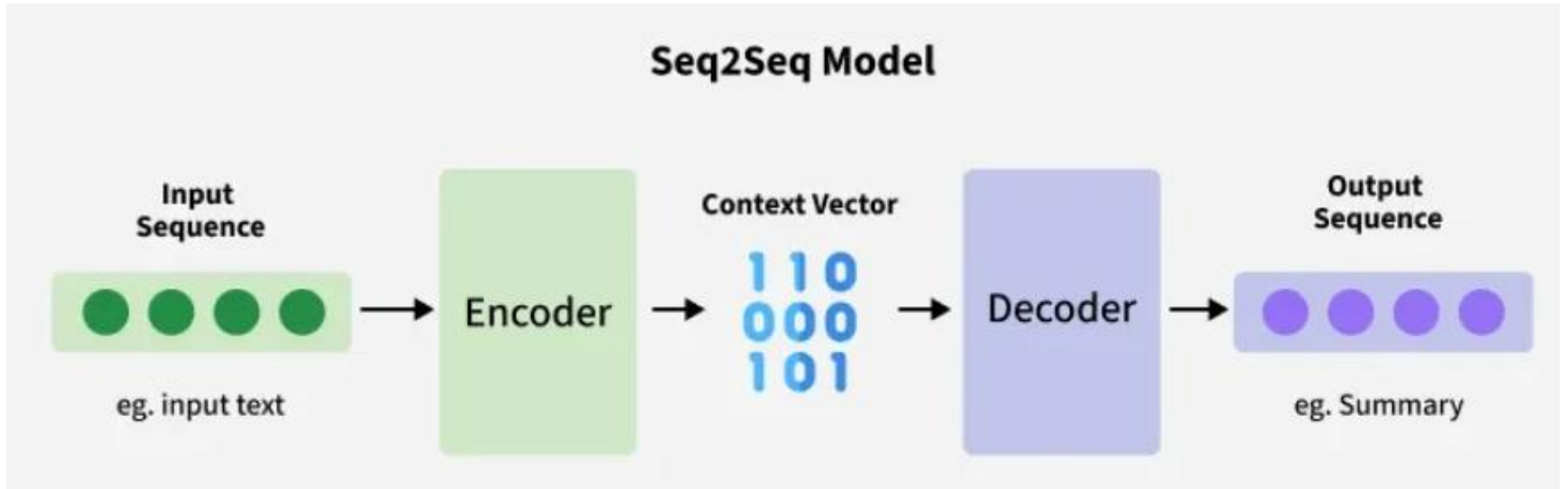


Figure: Sequence to Sequence Model Architecture

7.1 Sequence to Sequence Models

- The Sequence-to-Sequence Model architecture has two components:

1. The Encoder

- This is an RNN that reads the input sequence one token at a time.
- As it reads, it compresses the meaning of the entire input sentence into a single, fixed-length vector. We call this the context vector or the "thought vector." It represents the meaning of the sentence.

2. The Decoder

- This is another RNN.
- It takes that context vector as its initial hidden state and starts generating the output sequence token by token.
- We can think of it like a translator listening to a speech in English, taking notes (the context vector), and then delivering the speech in Nepali. The encoder is listening; the decoder is speaking.

7.1 Sequence to Sequence Models

Teacher Forcing

- During training, instead of feeding the decoder's own prediction back as the next input, we provide the correct previous word from the training data.
- Example: Correct sentence: "I love AI."
- During training: Decoder receives: "I" instead of its own predicted word.
- This technique is called Teacher Forcing and helps the model learn faster and more accurately.

7.1 Sequence to Sequence Models

The Bottleneck and The Attention Mechanism

- For a long time, sequence to sequence model worked, but it had a fatal flaw. If the input sentence is 100 words long, the encoder is forced to cram all that information into a single, fixed vector. The decoder, trying to generate the 50th output word, has to rely on that one vector. The information gets diluted.
- This is where Attention revolutionized the field (introduced in Bahdanau et al., 2015).
- Attention solves this bottleneck. Instead of passing only the final hidden state to the decoder, we pass all the hidden states from the encoder. Now, at each step of generating a new word, the decoder looks back at all the encoder's hidden states and asks: "Which part of the input sentence is most relevant to what I am trying to say right now?"
- It calculates a weighted sum of the encoder's states effectively "attending" to the most relevant part of the input.
- Example: If Input is "I am studying Artificial Intelligence." Then, when generating "Artificial" the decoder mainly attends to "Artificial Intelligence." This greatly improves translation quality.

7.1 Sequence to Sequence Models

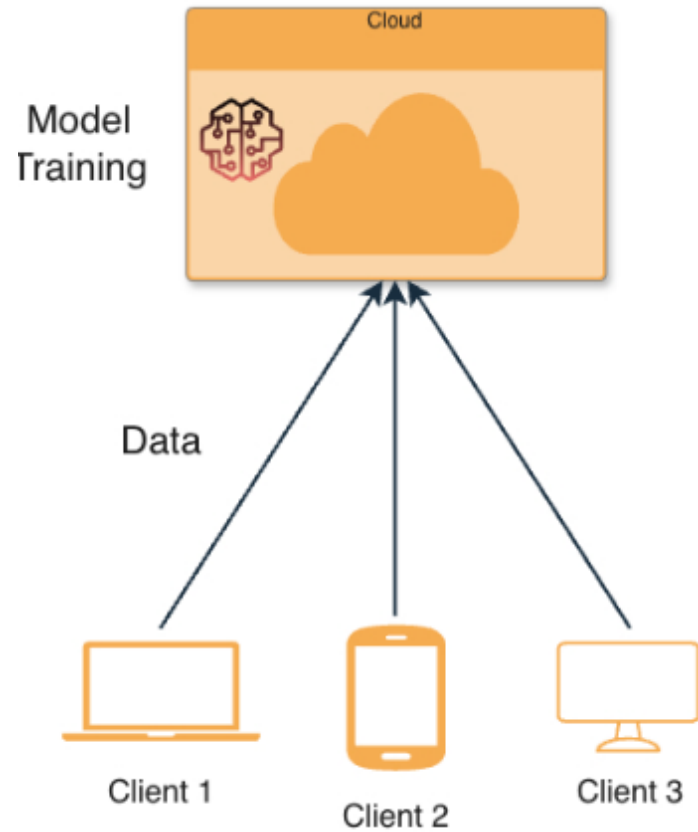
- Applications:
 - Machine Translation
 - Text Summarization
 - Speech Recognition
 - Chatbots
 - Code Generation
 - Image Captioning

7.2 Federated Learning

- Federated learning is a decentralized approach to training machine learning (ML) models.
- Each node across a distributed network trains a global model using its local data, with a central server aggregating node updates to improve the global model.
- Imagine we have 1 million smartphones. We want to train a next-word prediction keyboard. The traditional way is illegal (or immoral) because it would require uploading everyone's keystrokes to Google.
- Federated Learning offers a different approach

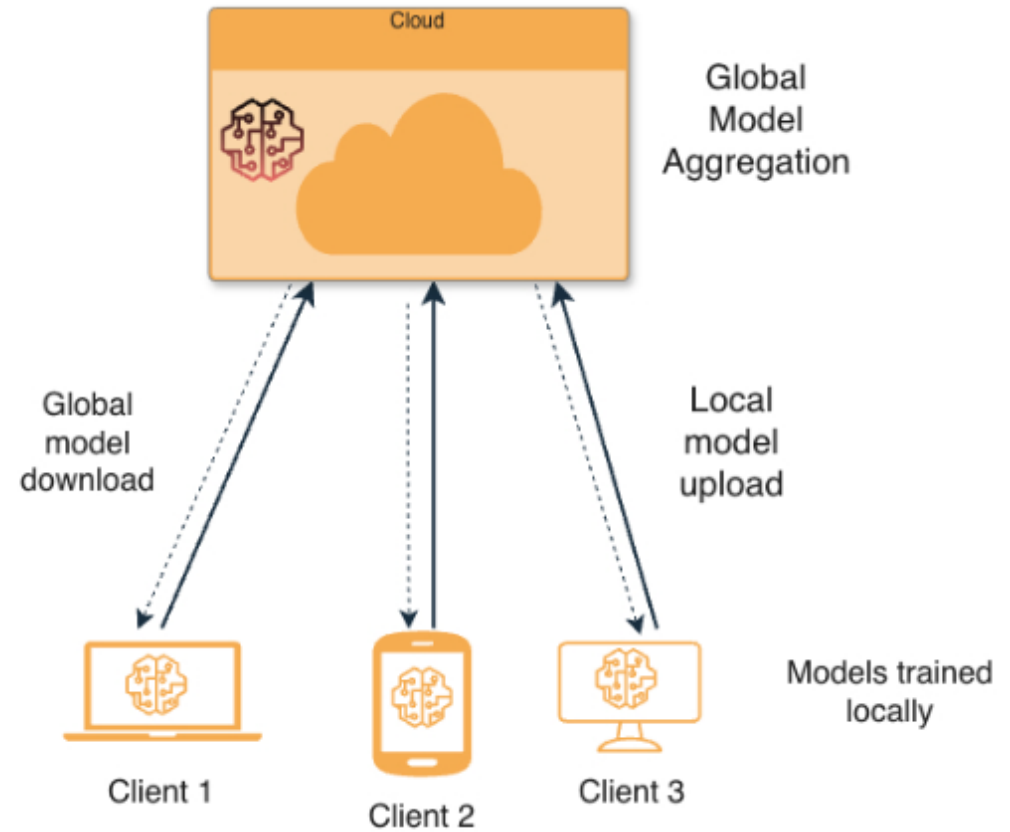
7.2 Federated Learning

Centralised Machine Learning



VS.

Federated Learning



7.2 Federated Learning

- Federated learning involves following stages:
 1. Initialization: The cloud server creates a global model with random weights.
 2. Selection: The server selects a random subset of devices (e.g., 10,000 phones).
 3. Broadcast: The server sends the current global model weights to these selected devices.
 4. Local Training: Each device downloads the model. It trains this model exclusively on the local data stored on that specific device (example: your text messages, your notes). The data never leaves the device. The device computes an update to the weights.
 5. Aggregation: The device send the weights (not the data) back to the server. The server uses an algorithm called Federated Averaging (FedAvg) to combine all these updates into a new, improved global model.
 6. Repeat: The process iterates.

7.2 Federated Learning

Benefits:

- Privacy: The raw data never leaves the device. It drastically reduces the risk of data leakage.
- Security: Even if a malicious actor intercepts the "weights," it is computationally infeasible to reverse-engineer the raw data from a gradient update.
- Data Naturalization: The data stays distributed, reflecting the natural distribution of usage. We get a model that works well across diverse use cases.

7.2 Federated Learning

The Challenges:

- Communication Costs: Sending weights back and forth over networks is expensive.
- Statistical Heterogeneity: The data on one device is not representative of the data on other devices. Users generate very different data distributions. Example: One might type in English; other might type in German. This "non-IID" (Non-Independent and Identically Distributed) data makes convergence difficult.
- Stragglers: Some devices are slow or have low battery. We need to wait for them, or we drop them, affecting convergence.

7.2 Federated Learning

Applications:

- Smartphone Keyboards
- Healthcare
- Banking
- Smartwatches
- Autonomous vehicles
- IoT devices

7.3 Edge AI

- Edge AI means running AI models directly on edge devices instead of relying on a remote cloud server.
- Examples of edge devices include: Smartphones, Drones, Security cameras, Robots, Smart speakers, Wearable devices, Industrial sensors.
- In Edge AI sensor collects data and sends it to the local AI model running in the edge device that does the prediction and takes immediate action. Therefore, no cloud is needed.
- Examples:
 - Smart Doorbell that recognizes visitors locally.
 - Smart Camera that detects intruders instantly.
 - Smartwatch that detects abnormal heart rate without internet.
 - Factory Robot that inspects defective products in real time.
 - Agricultural Drone that identifies crop diseases while flying over fields.

7.3 Edge AI

Benefits:

- **Low Latency:** Real-time inference. The model runs locally.
- **Privacy:** If the inference happens locally, we don't need to send the data to the cloud to get a result.
- **Reliability:** If the internet connection drops, the device still works.
- **Bandwidth Costs:** Processing data locally reduces the need to stream huge amounts of video or audio to the cloud.

7.3 Edge AI

The Challenge of Edge AI:

- Limited compute power
- Limited memory (RAM)
- Limited battery life

7.3 Edge AI

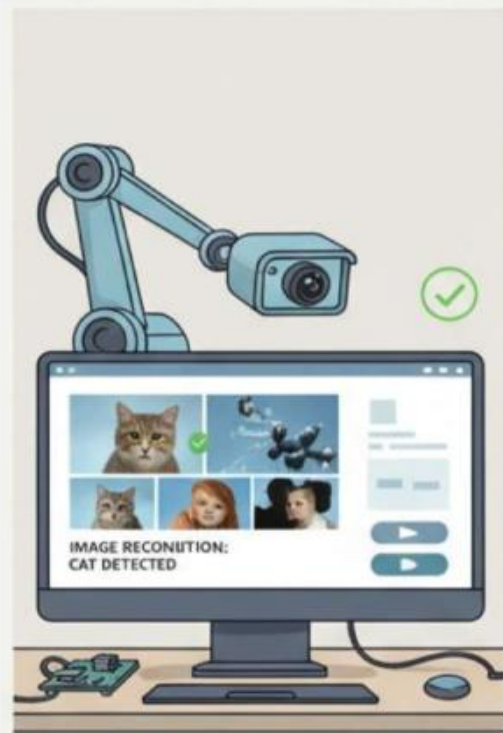
Cloud AI	Edge AI
Processing occurs in cloud	Processing occurs on device
Requires internet	Can work offline
Higher latency	Low latency
Better for heavy computation	Better for real-time decisions
Higher privacy concerns	Better privacy

7.4 Ethics and AI: Responsible AI

- In the era of AI, there are immense opportunities for socio-economic development both in Nepal and globally.
- However, rise of AI also presents risks including job displacement, vulnerabilities in national cybersecurity, elections manipulation, AI driven warfare and the disruption of national ethics.
- AI Ethics refers to the moral principles and guidelines that ensure Artificial Intelligence is designed, developed, and used in ways that benefit individuals and society while minimizing harm.
- AI Ethics ensures that AI is developed and used responsibly, fairly, transparently, and safely.

7.4 Ethics and AI: Responsible AI

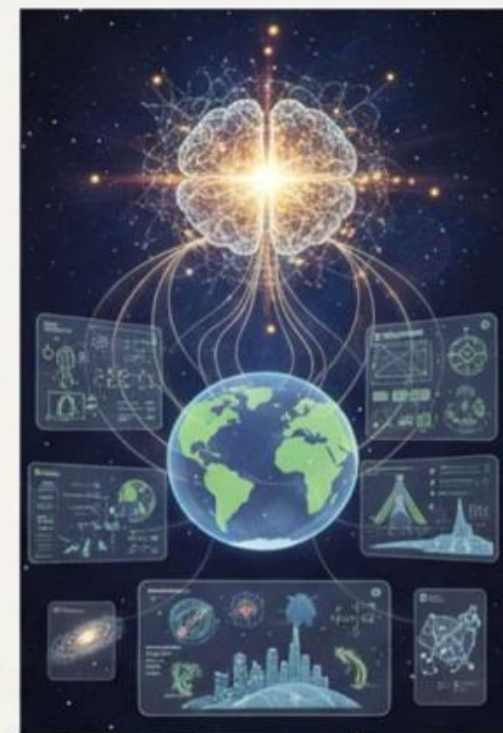
Types of AI



**Artificial Narrow
Intelligence (ANI)**



**Artificial General
Intelligence (AGI)**



**Artificial Super
Intelligence (ASI)**

7.4 Ethics and AI: Responsible AI

Threats of AI

AI is a “social weapon of mass destruction” to humanity

By Yuval Noah Harari, a renowned historian and author

- 1 Job Displacement
- 2 Surveillance and Privacy
- 3 Manipulation and Propaganda
- 4 Weaponization of AI
- 5 Existential Risks



7.4 Ethics and AI: Responsible AI

AI Risks in Nepal:

- Data Privacy Risks
- National Security Risks
- Economic Risks
- Legal and Regulatory Challenges
- Cultural and Social Risks
- Compliance and Governance Risks
- Weak Cybersecurity Practices
- Job displacement
- We are hanging in AI tools and our privacy is dumping in silicon valley.

7.4 Ethics and AI: Responsible AI

Workforce Displacement

- Generative AI could automate 60 to 70% of workloads.
- By the mid-2030s, up to 30% of jobs could be automatable.



 Teacher	 Customer Service Repres...	 Graphic Designers
 Law	 Paralegal	 Customer service
 Manufacturing	 Market Research Analysts	 Sales

Figure: The Top 9 Job Most at Risk of Being Replaced by AI

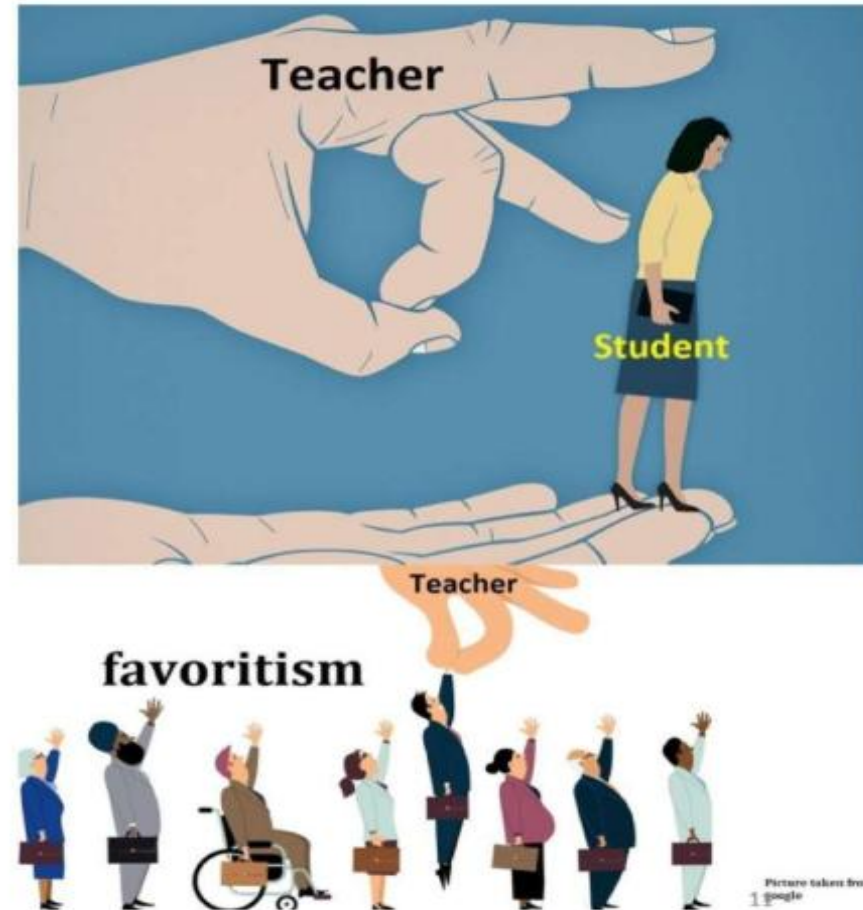
7.4 Ethics and AI: Responsible AI

Fairness of AI use

Makes fair decisions across different demographic groups.

Fairness in AI can be introduced through different **pre, post and in-processing methods**.

PRE-PROCESSING TECHNIQUES	IN-PROCESSING TECHNIQUES	POST-PROCESSING TECHNIQUES
Representative data collection and bias removal.	Regularization techniques and selection of best model.	Experimentation with the holdout set.



7.4 Ethics and AI: Responsible AI

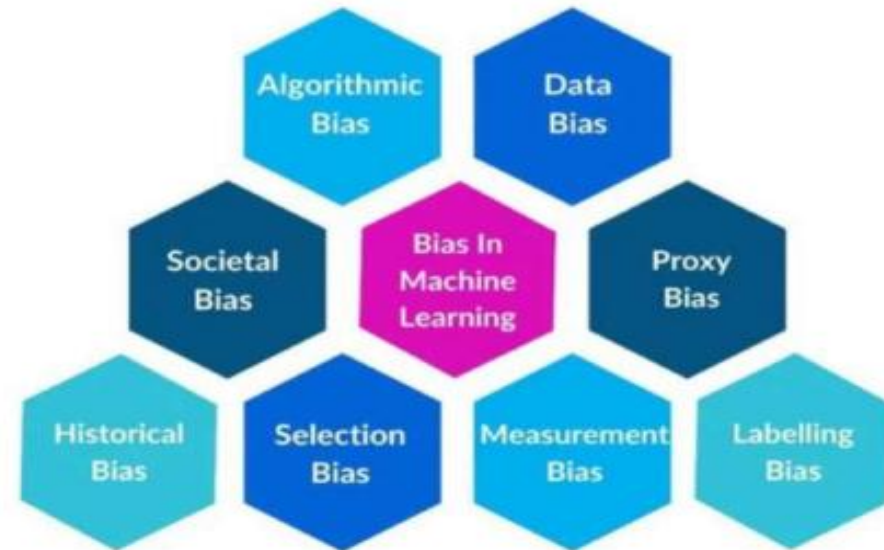
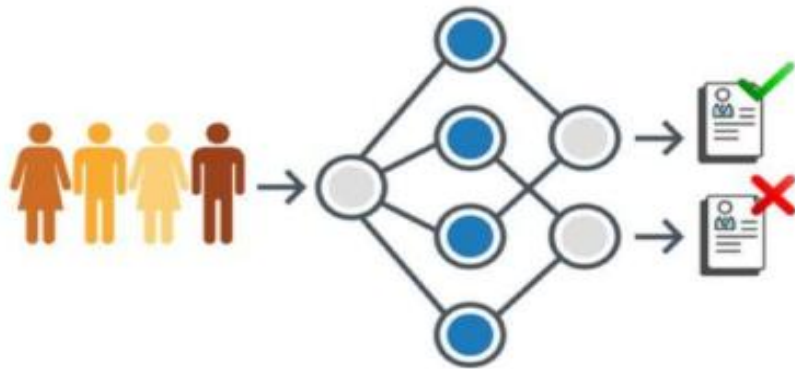
The main objective of fairness is to build a model that provides or makes fair decisions across **different demographic groups**



7.4 Ethics and AI: Responsible AI

Bias in AI System

- Bias in machine learning can manifest in various forms, influencing the outcomes of ML models and potentially leading to unfair or discriminatory results.



7.4 Ethics and AI: Responsible AI

Example: Language models of today are also biased on the basis of gender, race, ethnicity, etc.

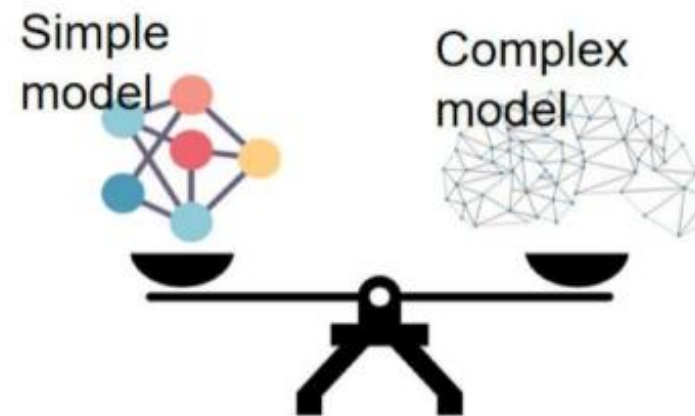


7.4 Ethics and AI: Responsible AI

The biases in the models seen today is a result of two things: **data bias** and **selection of models**.



- Sample size disparity
- Skewed/tainted data(Prejudice in the labels)
- Biased features

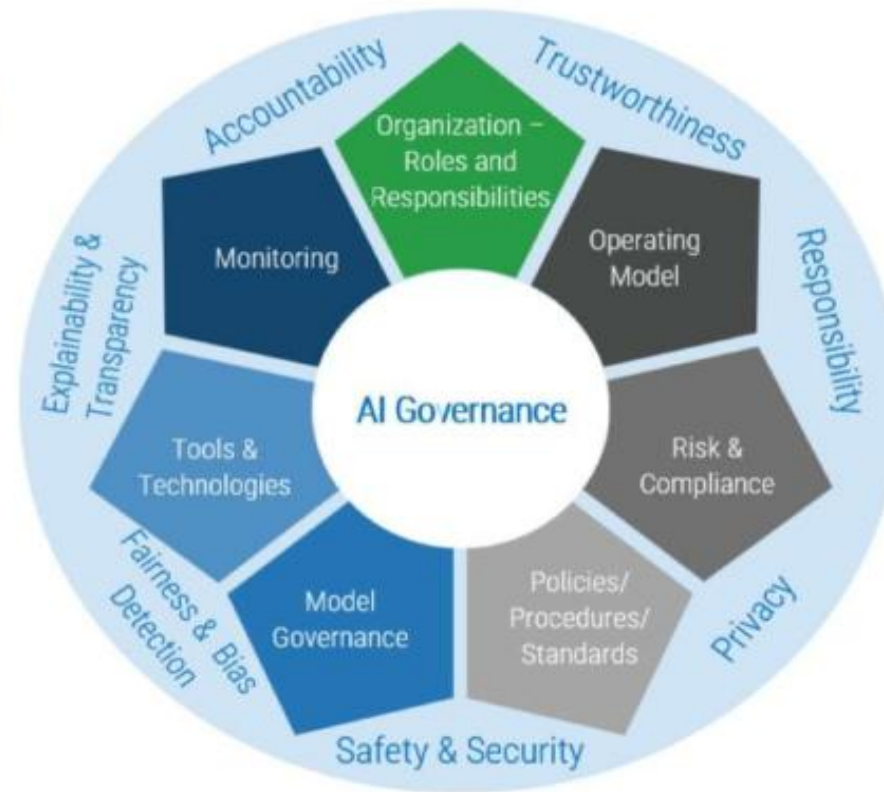


Models must be selected as per the problems complexity.

7.4 Ethics and AI: Responsible AI

AI Governance

- **AI governance** refers to the rules, policies, frameworks, and practices that guide the ethical, safe, and responsible development, deployment, and use of artificial intelligence systems, ensuring they align with human values, legal standards (like the EU AI Act), and organizational goals while mitigating risks like bias, privacy breaches, and misuse.
- It establishes transparency, accountability, and fairness as core principles, requiring ongoing monitoring and adaptation as AI technology evolves rapidly.



Responsible AI prospective

7.4 Ethics and AI: Responsible AI

Ethical AI: Key Considerations

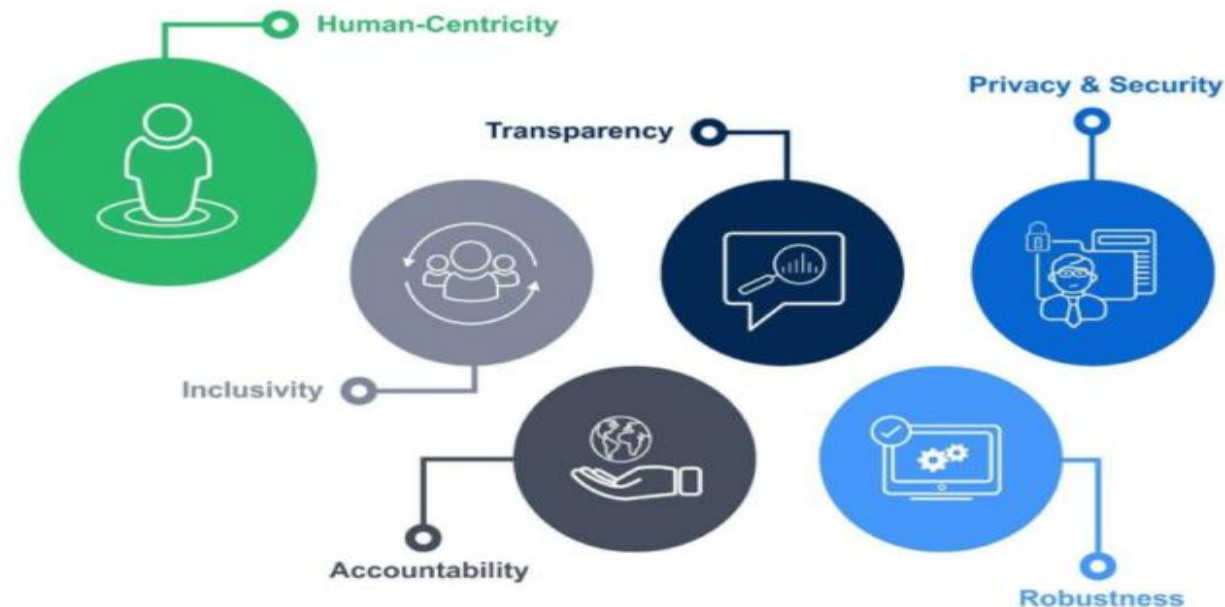
1. Fairness and Bias: Example: A hiring AI could accidentally favor one group over another.
2. Privacy: AI often uses large amounts of personal data. The system need to protect user privacy and keep data safe.
3. Transparency and Explainability: AI decisions can sometimes feel like a "black box." People should understand why an AI gave a certain output-especially in health, education, or law.
4. Accountability: If something goes wrong, who is responsible? Make sure developers and companies take responsibility for AI behavior.
5. Safety and Security: AI systems must be protected from failures or misuse. The issues could be cyber-attacks, system malfunction and AI spreading misinformation.
6. Impact on Jobs and Economy: While some jobs may evolve or disappear due to AI, it will simultaneously generate new opportunities, necessitating upskilling and reskilling to equip people with relevant abilities.
7. Human Control and Autonomy: Humans should stay in control of important decisions. AI should support humans, not replace critical judgment.
8. Social and Cultural Impact: AI can shape how we learn, communicate, and interact online. Use AI to support healthy and positive social development.
9. Environmental Impact: Training large AI models uses energy. Therefore, develop eco- friendly and energy-efficient AI systems.

7.4 Ethics and AI: Responsible AI

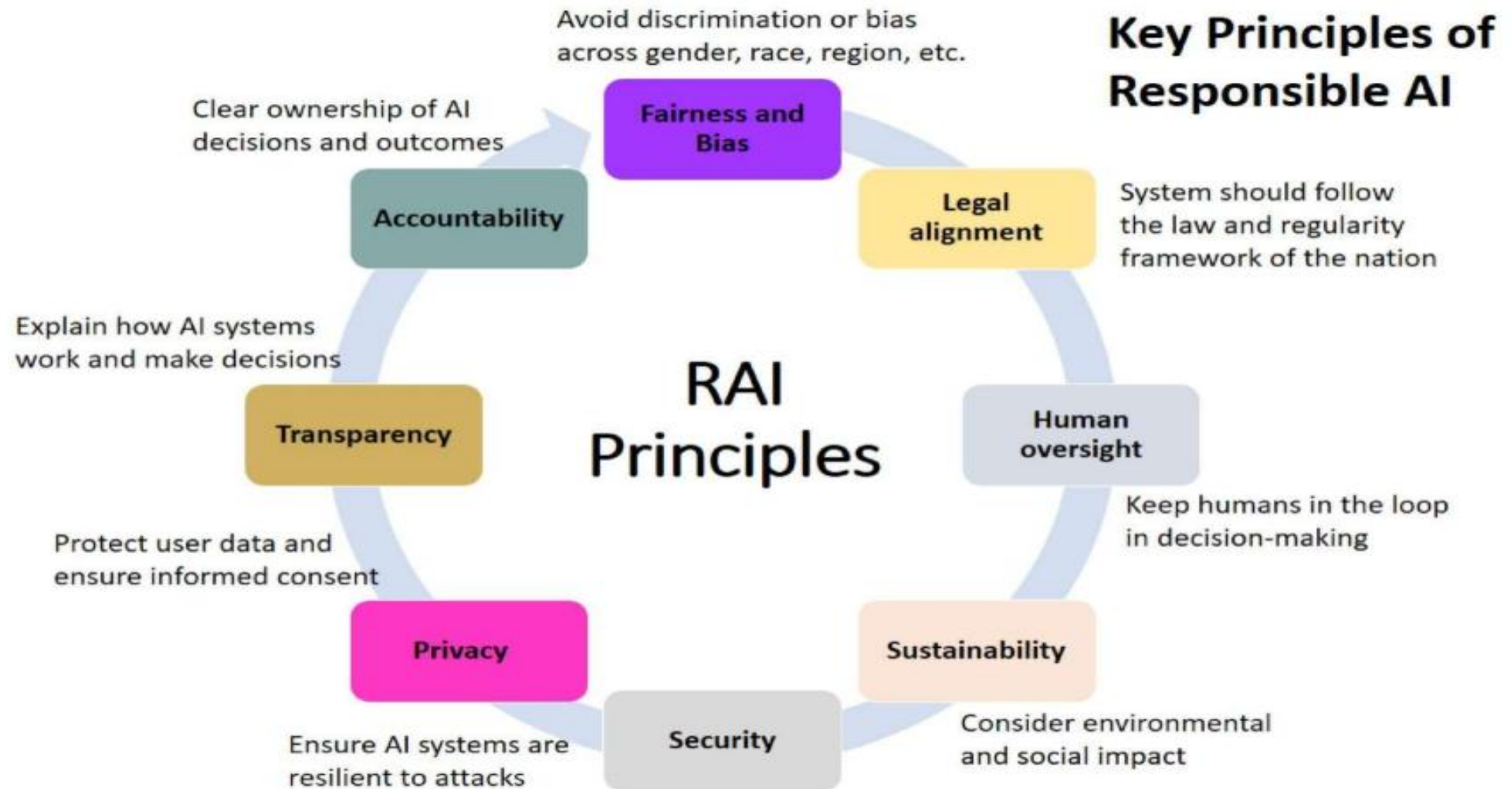
Responsible AI (RAI):

- Responsible AI refers to design, development, deployment and use of AI systems in a way that is ethical, transparent, fair, accountable, lawful, safe, beneficial and aligned with human values, societal norms and legal frameworks.

Responsible AI (RAI)



7.4 Ethics and AI: Responsible AI



7.4 Ethics and AI: Responsible AI

- Implementing Responsible AI in Organization



7.4 Ethics and AI: Responsible AI

UNESCO AI Ethics:

- UNESCO adopted the Recommendation on the Ethics of AI in November 2021.
- This was the first global standard-setting instrument on AI ethics, agreed upon by its 193 Member States.
- Core Values:
 1. Respect, protection, and promotion of human rights and fundamental freedoms and human dignity: This is the cornerstone, ensuring AI development and use uphold universal human rights.
 2. Environmental and ecosystem flourishing: AI should be developed and used in a way that contributes to environmental sustainability and protects ecosystems.
 3. Ensuring diversity and inclusiveness: AI should foster diversity in all its forms (cultural, social, gender, linguistic, etc.) and promote inclusion, ensuring benefits are accessible to all.
 4. Living in peaceful, just, and interconnected societies: AI should contribute to building equitable, peaceful, and interconnected societies, fostering social well-being and justice.

7.4 Ethics and AI: Responsible AI

A human rights approach to AI

- Ten core principles lay out a human-rights centered approach to the Ethics of AI.
 1. Proportionality and Do No Harm: The use of AI systems must not go beyond what is necessary to achieve a legitimate aim (Risk assessment}
 2. Safety and Security: Vulnerabilities to attack (security risks) should be avoided and addressed by AI actors.
 3. Right to Privacy and Data Protection: Privacy must be protected and promoted throughout the AI lifecycle. Adequate data protection frameworks should also be established.
 4. Multi-stakeholder and Adaptive Governance & Collaboration: The development of AI policies and practices should involve diverse stakeholders (governments, civil society, academia, private sector) and adapt to the rapid pace of technological change. International law and national sovereignty must be respected.
 5. Responsibility and Accountability: AI systems should be auditable and traceable. Follow human rights norms and threats to environmental wellbeing.

7.4 Ethics and AI: Responsible AI

6. Transparency and Explainability: The ethical deployment of AI systems depends on their transparency & explainability (T&E).
7. Human Oversight and Determination: Humans must retain the ability to oversee, intervene in, and ultimately override decisions made by AI systems, especially in critical contexts.
8. Sustainability: AI technologies should be assessed for their impacts on environmental sustainability (e.g., energy consumption, resource use) and contribute to achieving the UN's Sustainable Development Goals.
9. Awareness & Literacy: Public understanding of AI and data should be promoted through open & accessible education, civic engagement, digital skills & AI ethics training, media & information literacy.
10. Fairness and Non-Discrimination: AI actors should promote social justice, fairness, and non-discrimination while taking an inclusive approach to ensure AI's benefits are accessible to all.

7.4 Ethics and AI: Responsible AI

EU Guidelines for Trustworthy AI

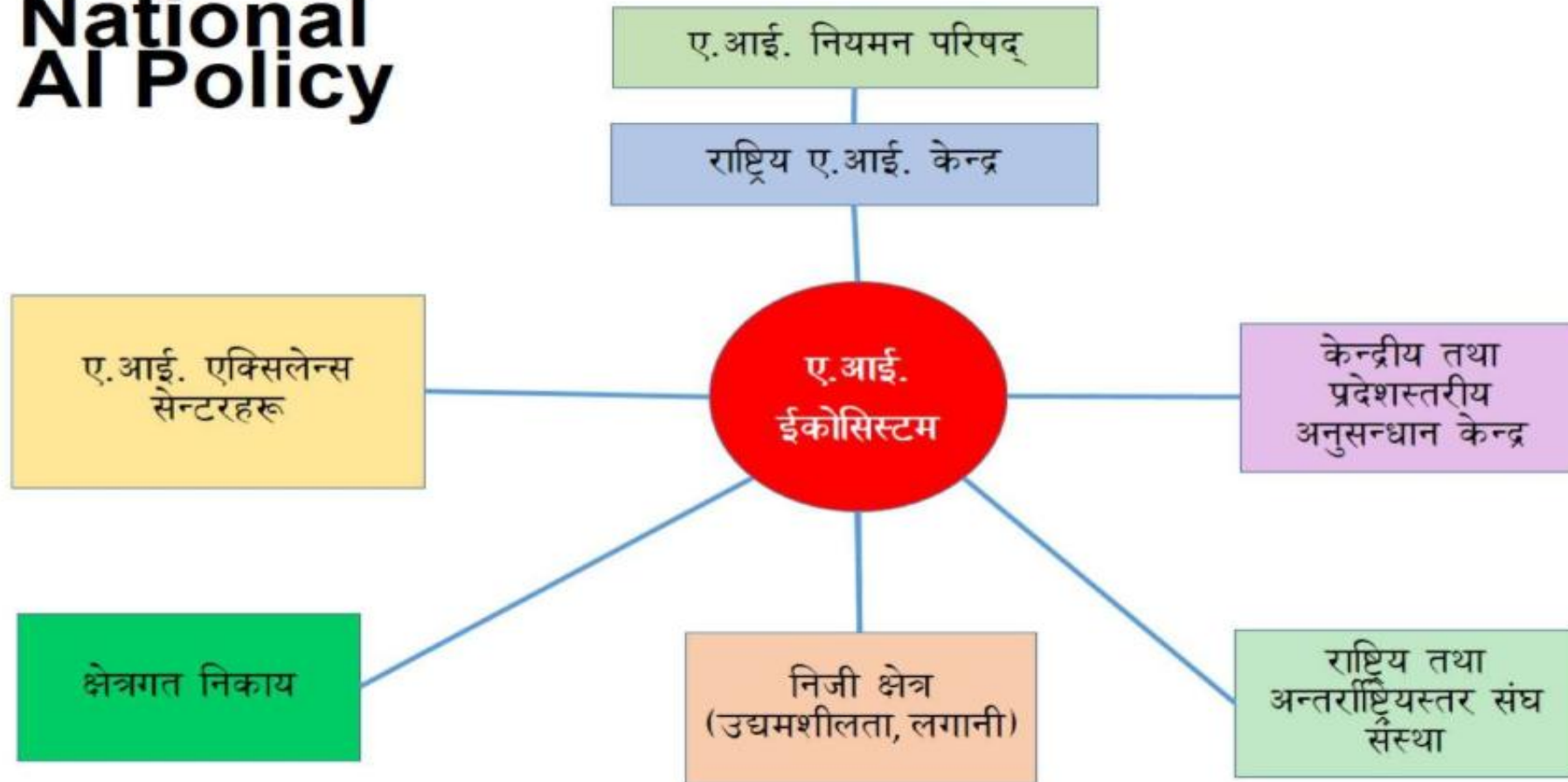
- The EU Guidelines for Trustworthy AI are built around a framework of three interconnected components and seven key requirements that AI systems should meet .
- To be considered trustworthy, an AI system must be:
- Lawful: Respecting all applicable laws and regulations .
- Ethical: Adhering to ethical principles and values .
- Robust: Being technically reliable and secure, considering its social environment

7.4 Ethics and AI: Responsible AI

- To put these components into practice, the Guidelines define seven specific requirements :
- Human Agency and Oversight: AI systems should empower humans and allow for informed decision-making. Proper oversight mechanisms, such as human-in-the-loop approaches, must be ensured .
- Technical Robustness and Safety: AI systems need to be resilient, secure, and accurate to prevent and minimize unintentional harm .
- Privacy and Data Governance: There must be full respect for privacy and data protection, alongside robust data governance mechanisms that ensure data quality and integrity .
- Transparency: The data, system, and business models should be transparent. Traceability and the ability to explain an AI system's decisions are crucial .
- Diversity, Non-discrimination, and Fairness: Unfair bias must be avoided to prevent marginalization or discrimination. AI systems should be accessible to all stakeholders .
- Environmental and Societal Well-being: AI systems should benefit all people and be sustainable and environmentally friendly, with careful consideration of their social impact .
- Accountability: Mechanisms should be in place to ensure responsibility for AI systems and their outcomes. Auditability and redress mechanisms play a key role.

7.4 Ethics and AI: Responsible AI

National AI Policy



References

- Russell, S., Norvig, P. (2021). Artificial intelligence: A modern approach. Pearson.